

The Five Assumptions of Linear Regression

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Multicollinearity ([Note 27](#)); Q-Q plot ([Note 30](#)).

1. Overview

Key point: Linear regression is reliable when five conditions hold: a linear relationship, no multicollinearity, normal residuals, constant spread of residuals, and no pattern in the residuals over the rows.

Linear regression always produces a line or hyperplane, whatever the data looks like. Whether that line can be trusted depends on a few conditions, called the **assumptions** of linear regression. Like the rules of a game, they do not stop anyone from playing, but break them and the score means little.

Textbooks list them in slightly different ways; these five are the core ones (ISL §3.3.3; Kutner Ch. 3 and §7.6):

	Assumption	Checked with
1	Linear relationship between each feature and the output	scatter plots
2	No multicollinearity between the features	VIF, correlation heatmap
3	Residuals are normally distributed	histogram, Q-Q plot
4	Homoscedasticity: residuals have the same spread everywhere	residuals against predictions
5	No autocorrelation of the residuals	residuals in row order

Three words first. A **feature** is an input variable (one column of the data table), an **observation** is one record (one row), and the **target** is the output we predict.

The first two assumptions concern the features; the last three concern the **residuals**, the errors $y_i - \hat{y}_i$ on each observation.

The example data has 200 observations, three features and one target. A linear regression trained on 70% of it scores $R^2 = 0.96$ on the other 30% (60 observations); each check below uses those 60 test residuals.

Python: The model whose assumptions we check.

```
df = pd.read_csv("data/data.csv")
X, y = df.iloc[:, :3].to_numpy(), df["target"].to_numpy()
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)
model = LinearRegression().fit(X_train, y_train)
residual = y_test - model.predict(X_test)
```

2. Assumption 1: a linear relationship

Key point: Each feature should relate to the output along a straight line, rising or falling. A curved relationship needs a different model.

Linear regression fits straight lines, so the relationship between each feature and the output should be roughly linear. Both directions are fine:

- **positive:** the output rises as the feature rises;
- **negative:** the output falls as the feature rises.

A curved relationship (for example, the output growing with the square of the feature) breaks this assumption, because a straight line then misses the pattern systematically.

The check is a scatter plot of the output against each feature (Figure 1).

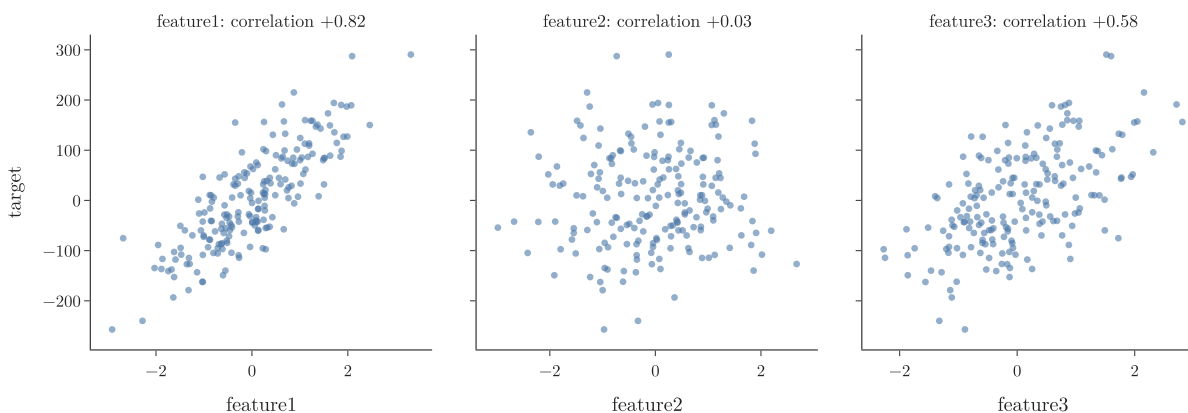


Figure 1: Each feature against the target

- **feature1:** a clear rising line (correlation 0.82).
- **feature3:** also rising, with more scatter (0.58).
- **feature2:** no visible relationship at all (0.03). Its fitted coefficient is only -0.28 , against 72.7 and 53.3 for the other two: the model has learned to almost ignore it.

Extra: A feature with no relationship, like feature2, does not break the assumption so much as add nothing; the model gives it a near-zero weight. When a feature has a curved relationship, a simple approach is to add non-linear transformations of that feature, such

as $\log x$, $\sqrt{x}$ or x^2 , to the model (ISL §3.3.3). Adding x^2 is polynomial regression, a later Note.

3. Assumption 2: no multicollinearity

Key point: The features should not be strongly related to each other. If they are, the model cannot tell which one deserves the credit, and the coefficients become unreliable.

Multicollinearity means one feature can be (largely) predicted from the others. Multicollinearity was met in the one-hot encoding Note with the dummy variable trap.

3.1 Why it is a problem

Key point: Each coefficient measures the effect of its feature with the others held fixed. Related features cannot be held fixed separately.

A coefficient β_1 is the change in the output when x_1 rises by 1 *and all other features stay the same*. If x_1 and x_3 always move together, “change x_1 and keep x_3 fixed” never happens in the data. The model then cannot separate their effects.

An analogy: two scientists finish a project together. If one is a physicist and the other a chemist, it is easy to say who contributed what. If both have identical skills, it is impossible. With related features the model faces the second case, so the coefficients become unreliable: a small change in the data can move them a long way (ISL §3.3.3). The predictions, on the other hand, usually stay good (Kutner §7.6).

3.2 Checking it

Key point: A VIF near 1 means no multicollinearity; above 5 is a problem. All three features here have VIF 1.01.

The standard check is the **variance inflation factor (VIF)**. For each feature, a regression predicts that feature from all the other features; if they predict it well (high R^2), the feature is redundant.

In words: the VIF of a feature is one divided by the part of it the other features cannot explain.

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 of predicting feature j from the others. With numbers: if the others explain 80% of a feature ($R_j^2 = 0.8$), its VIF is $1/0.2 = 5$, the usual danger line. If they explain none of it, the VIF is 1.

Python: VIF with statsmodels.

```
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)
from statsmodels.tools import add_constant

Xc = add_constant(X_train)      # add an intercept column
[variance_inflation_factor(Xc, i) for i in range(1, 4)]
# [1.011, 1.010, 1.014]
```

A quicker, rougher check is a heatmap of the correlations between the features (Figure 2). Here every correlation between different features is at most 0.06: no multicollinearity.

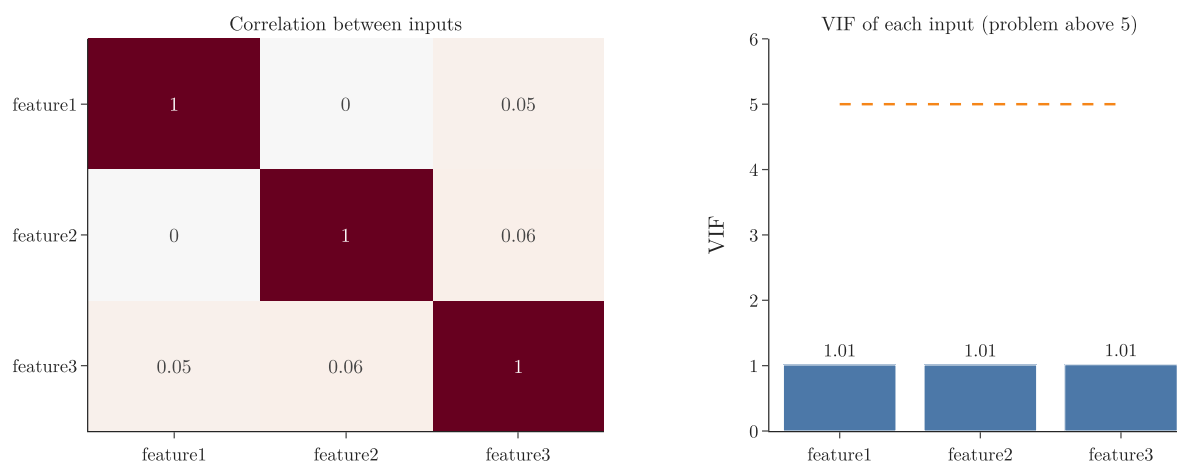


Figure 2: Correlations between the features, and their VIF

Extra: If VIF is high, two simple fixes are to drop one of the related features, or to combine them into one feature, such as their average after scaling (ISL §3.3.3). Ridge regression, a later Note, was also designed for related features (Hoerl and Kennard).

4. Assumption 3: normal residuals

Key point: The residuals should follow a bell-shaped (normal) distribution centred on 0: most errors small, few large, positive and negative alike.

When the model is right on average, its errors should scatter around 0: many small errors and few large ones, as often too high as too low. Such a bell-shaped spread is a normal distribution.

Two checks, both on the residuals (Figure 3, top):

- **Histogram or density plot:** roughly a bell centred on 0. Here it is, apart from a small bump.

- **Q-Q plot** (from the function `transformer.Note`): the points should lie on the straight line. Here they do, with small wiggles.

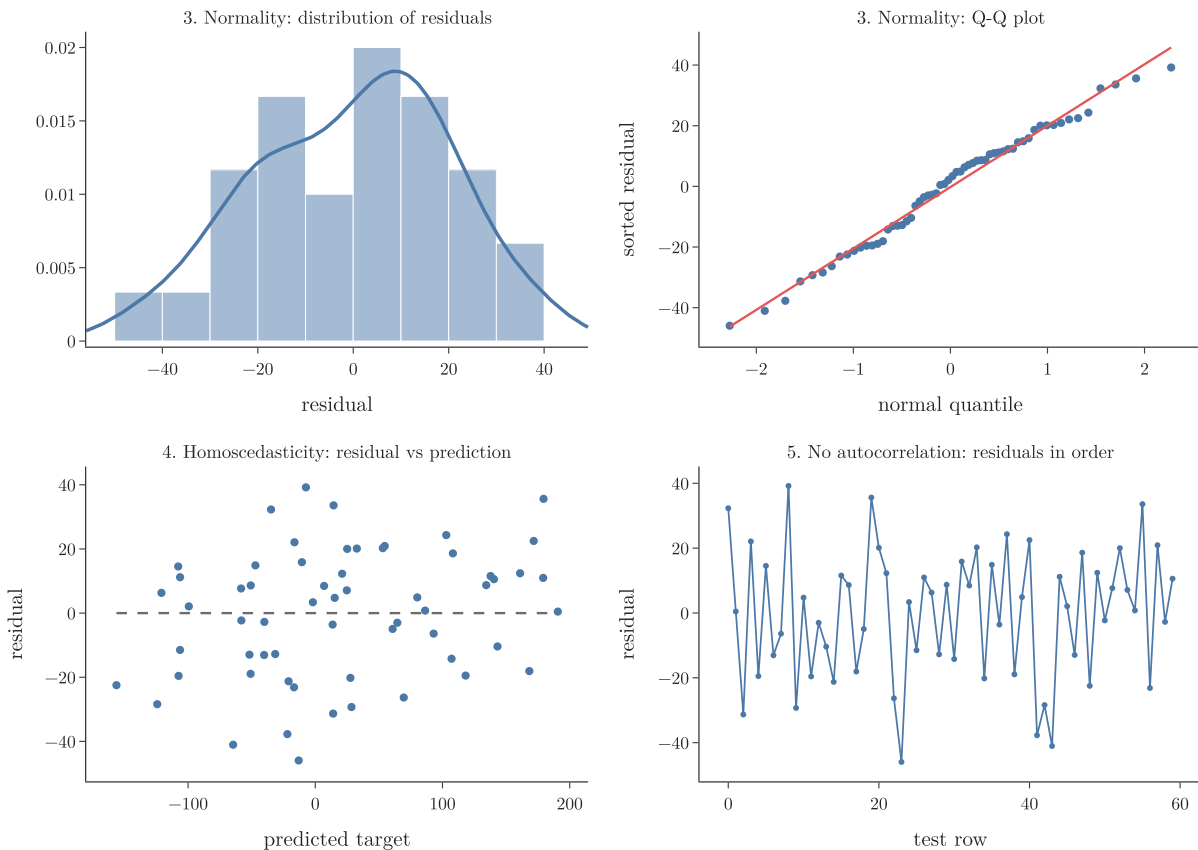


Figure 3: Residual checks for assumptions 3, 4 and 5

Extra: Formal tests exist as well. The **Shapiro-Wilk test** gives a p-value; above 0.05 means no evidence against normality. Here $p = 0.51$, and the skewness of the residuals is -0.23 , close to 0. With large samples these tests flag even small departures from normality, so look at the plots too (Ghasemi and Zahediasl 2012).

5. Assumption 4: homoscedasticity

Key point: The residuals should have the same spread for small and large predictions. A funnel shape (heteroscedasticity) breaks the assumption.

Homoscedasticity means “same scatter”: the size of the errors does not depend on the size of the prediction. Its opposite, **heteroscedasticity**, is common (ISL §3.3.3): for example, a house-price model that is off by a few thousand on cheap houses and by much more on expensive ones.

The check is a scatter plot of residuals against predicted values (Figure 3, bottom left). The

plot should look like an even band around 0, with no shape. Here it does: the spread is roughly the same from the lowest to the highest prediction.

Figure 4 (middle) shows what heteroscedasticity looks like: a funnel that widens to the right.

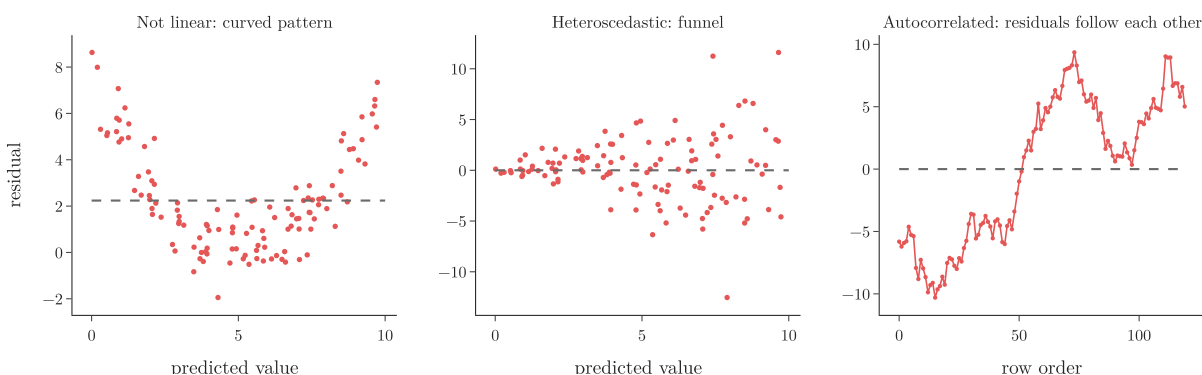


Figure 4: What failed assumptions look like (made-up data)

Extra: Heteroscedasticity does not make the coefficients wrong on average, but it makes the model's uncertainty estimates wrong: the confidence intervals and p-values that statistics packages report (Wooldridge §8.1). One fix is to take the log or square root of the target. The log shrinks large values more than small ones, and can turn a funnel into an even band (ISL §3.3.3).

6. Assumption 5: no autocorrelation of the residuals

Key point: One residual should not predict the next. Plotted in row order, the residuals should jump around randomly, not drift in waves.

Autocorrelation means each residual is related to the one before it: if the model was too high on one observation, it is also too high on the next. Autocorrelation is common in time-series data, measurements taken at points in time (ISL §3.3.3).

The check is to plot the residuals in row order (Figure 3, bottom right). They should jump up and down with no pattern. Here they do.

Figure 4 (right) shows positive autocorrelation: long runs above 0 followed by long runs below, like a slow wave. One cause is a missing feature that changes slowly from one observation to the next, such as a time trend: the model cannot see the trend, so the trend shows up in the residuals (tested in the Extra below). Autocorrelation makes the model look more certain than it is: its confidence intervals come out too narrow (ISL §3.3.3).

Extra: The **Durbin-Watson statistic** puts a number on autocorrelation: about 2 means no autocorrelation, towards 0 means positive autocorrelation, towards 4 negative (statsmodels docs). Here it is 2.31, close to 2: no sign of autocorrelation. To test the

missing-trend cause, the notebook makes data $y = 2x + 3 \sin(t/25) + \text{noise}$, with t the observation number. Fitting on x alone gives Durbin-Watson 0.40; adding the slow wave $\sin(t/25)$ as a feature brings it back to 2.09. Figure 4 (left) also shows a residual plot when assumption 1 fails: a curve instead of a flat band.

7. Summary

Assumption	What it means	How to check	This data
1. Linearity	each feature relates to the output along a line	scatter plots	yes (feature2 unrelated)
2. No multicollinearity	features not related to each other	VIF (problem above 5), correlation heatmap	VIF 1.01
3. Normal residuals	errors form a bell around 0	histogram, Q-Q plot	yes
4. Homoscedasticity	errors have the same spread everywhere	residuals vs predictions	yes
5. No autocorrelation	errors do not follow each other	residuals in row order	yes

- Assumptions 1 and 2 are about the features; 3 to 5 are about the residuals.
- A failed assumption does not stop the model from running; it makes its coefficients or uncertainty estimates untrustworthy.
- Residual plots should look like random noise; any shape (curve, funnel, wave) points to a broken assumption.

8. Sources

- **ISL**: James, Witten, Hastie, Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021. §3.3.3, Potential Problems.
- **Kutner**: Kutner, Nachtsheim, Neter, Li, *Applied Linear Statistical Models*, 5th ed., McGraw-Hill, 2005. Ch. 3 (diagnostics and remedial measures), §7.6 (multicollinearity and its effects).
- **Hoerl and Kennard**: A. E. Hoerl and R. W. Kennard, “Ridge Regression: Biased Estimation for Nonorthogonal Problems”, *Technometrics* 12(1), 55–67, 1970.
- **Ghasemi and Zahediasl**: A. Ghasemi and S. Zahediasl, “Normality Tests for Statistical Analysis: A Guide for Non-Statisticians”, *International Journal of Endocrinology and Metabolism* 10(2), 486–489, 2012.
- **Wooldridge**: J. M. Wooldridge, *Introductory Econometrics: A Modern Approach*, Cengage. Chapter 8, Heteroskedasticity (§8.1, consequences for OLS).
- **statsmodels docs**: statsmodels documentation, `statsmodels.stats.stattools.durbin_watson`.

9. Key terms

Term	Meaning
Feature	An input variable: one column of the data table

Term	Meaning
Observation	One record: one row of the data table
Target	The output we predict
Assumption (of a model)	A condition the data must meet for the model's results to be reliable
Residual	The error on one data point: actual minus predicted value
Variance inflation factor (VIF)	$1/(1 - R_j^2)$: how well the other features predict feature j ; above 5 signals multicollinearity
Homoscedasticity	The residuals have the same spread for all predicted values
Heteroscedasticity	The spread of the residuals changes with the predicted value, often as a funnel
Autocorrelation	Each residual is related to the one before it in row order
Shapiro-Wilk test	A statistical test of whether data follows a normal distribution
Durbin-Watson statistic	A number from 0 to 4 measuring autocorrelation of residuals; about 2 means none
statsmodels	A Python library for statistical models and tests